

Math 262 – Solutions to Practice Problems

1. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix}.$$

- (a) Show that T is a linear transformation.
(b) Find the matrix of T .

Solution.

1.

$$\begin{aligned} a^{(*)} T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) &= T\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}\right) \\ &= \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} - \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} + \begin{bmatrix} x_2 + y_2 \\ x_3 + y_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} - \begin{bmatrix} x_1 + x_2 + y_1 + y_2 \\ x_2 + y_2 + x_3 + y_3 \end{bmatrix} = T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) + T\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right). \end{aligned}$$

- (i) For $r \in \mathbb{R}$:

$$T\left(r \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} rx_1 + rx_2 \\ rx_2 + rx_3 \end{bmatrix} = r \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix} = r T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right).$$

So T is linear.

2.

$$T(e_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

2. Given $A\vec{x} = \vec{b}$, suppose the rref of the augmented matrix is

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Give the general solution to the system.

Solution.

$$\begin{cases} x_1 + x_4 = 0 \\ x_2 + 2x_4 = 1 \\ x_3 + 3x_4 = 2 \end{cases} \Rightarrow \begin{cases} x_1 = -x_4 \\ x_2 = 2x_4 + 1 \\ x_3 = -3x_4 + 2 \end{cases} \quad (x_4 = r).$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -r \\ 2r + 1 \\ -3r + 2 \\ r \end{bmatrix}.$$

Problem 3 (Original Statement): Given $\begin{cases} x_1 - x_3 = 1, \\ 2x_1 - 2x_2 = 1, \\ 3x_1 - 4x_2 + x_3 = a, \end{cases}$ find (a) no solution,

(b) exactly one solution, (c) infinitely many solutions; give the general solution when infinite.

Solution.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ -2 & 2 & 0 & 1 \\ 3 & -4 & 1 & a \end{array}\right]$$

Row ops: $R_2 \rightarrow R_2 + 2R_1$, $R_3 \rightarrow R_3 - 3R_1$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & -2 & 2 & -1 \\ 0 & -4 & 4 & a - 3 \end{array}\right]$$

Row op: $R_2 \rightarrow (-\frac{1}{2})R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & -4 & 4 & a - 3 \end{array}\right]$$

Row op: $R_3 \rightarrow R_3 + 4R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & 0 & 0 & a - \frac{1}{2} \end{array}\right]$$

(a) No solution if $a \neq \frac{1}{2}$. (b) Never exactly one. (c) Infinitely many if $a = \frac{1}{2}$.

If $a = \frac{1}{2}$, let $x_3 = r$. Then $x_2 = \frac{1}{2} + r$, $x_1 = 1 + r$, so $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 + r \\ \frac{1}{2} + r \\ r \end{bmatrix}$.

In $\mathbb{R}^2$, let $\vec{u} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$.

- (a) Show that $\vec{u}$ is a unit vector.

- (b) Let L be the line spanned by $\vec{u}$, and let $\vec{x} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Find the reflection $\text{ref}_L(\vec{x})$.

Solution.

$$|\vec{u}| = \sqrt{\left(\frac{1}{\sqrt{10}}\right)^2 + \left(\frac{3}{\sqrt{10}}\right)^2} = 1, \quad \text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}.$$

$$\vec{x} \cdot \vec{u} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} = -\frac{1}{\sqrt{10}} + \frac{12}{\sqrt{10}} = \frac{11}{\sqrt{10}}, \quad \text{proj}_L(\vec{x}) = \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \end{bmatrix}.$$

$$\text{ref}_L(\vec{x}) = 2 \text{proj}_L(\vec{x}) - \vec{x}.$$

Original Problem: In $\mathbb{R}^3$, let L be the line spanned by $\vec{w} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$, $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Verify that

$\vec{x} - \text{proj}_L(\vec{x})$ is orthogonal to $\vec{w}$.

Solution.

$$2 \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \\ \frac{11}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} \\ \frac{66}{10} \\ \frac{22}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} + 1 \\ \frac{66}{10} - 4 \\ \frac{22}{10} \end{bmatrix} = \begin{bmatrix} \frac{16}{5} \\ \frac{1}{5} \\ \frac{11}{5} \end{bmatrix}.$$

Note: You could also use the matrix formula for this one.

$$\text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w} = \frac{2}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\vec{x} - \text{proj}_L(\vec{x}) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = 0, \quad \text{so it is orthogonal.}$$

Problem 6 (Original Statement): Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$. Assume T is linear.

- (a) Find A , the matrix of T .

- (b) Use A to explain why T is invertible.

- (c) Use A to find the rule for T^{-1} .

Solution.

$$T(e_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

$$\det A = (1)(1) - (-1)(1) = 2 \Rightarrow A \text{ invertible}, \quad T^{-1} = T_{A^{-1}}, \quad A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

$$T^{-1}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(x_1 + x_2) \\ \frac{1}{2}(-x_1 + x_2) \end{bmatrix}.$$

Problem 7 (Original Statement):

Let θ be any angle and let A be the matrix which represents the counterclockwise rotation through θ . Prove that for any vector

$$\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \quad |A\vec{u}| = |\vec{u}|$$

—that is, rotations preserve length.

Solution:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad A\vec{u} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

$$|A\vec{u}| = \sqrt{(x \cos \theta - y \sin \theta)^2 + (x \sin \theta + y \cos \theta)^2}$$

Expand:

$$= \sqrt{x^2 \cos^2 \theta - 2xy \cos \theta \sin \theta + y^2 \sin^2 \theta + x^2 \sin^2 \theta + 2xy \sin \theta \cos \theta + y^2 \cos^2 \theta}$$

$$= \sqrt{x^2 (\cos^2 \theta + \sin^2 \theta) + y^2 (\sin^2 \theta + \cos^2 \theta)} = \sqrt{x^2 + y^2} = |\vec{u}|.$$

Hence rotations preserve length.

Problem 8 (Original Statement): Fill in the blanks for an invertible $n \times n$ matrix A :

- (a) $\text{rank}(A) =$ _____

- (b) The homogeneous system $A\vec{x} = \vec{0}$ has only the _____ solution.

- (c) A is row-reducible to the _____ matrix.

Solution.

$\text{rank}(A) = n$, only the trivial solution, A row-reduces to the identity.

Let $A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{bmatrix}$. Give the reduced row echelon form of A .

Solution.

$$\left[\begin{array}{ccccc} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{array}\right]$$

Row ops: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$, $R_4 \rightarrow R_4 - 6R_1$.

$$\left[\begin{array}{ccccc} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 1 & 0 & -2 & 5 \\ 0 & 2 & 4 & -4 & 6 \end{array}\right]$$

Row ops: $R_3 \rightarrow R_3 - R_2$, $R_4 \rightarrow R_4 - 2R_2$.

$$\left[\begin{array}{ccccc} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 0 & -4 & 0 & 4 \\ 0 & 0 & -4 & 0 & 4 \end{array}\right]$$

Row ops: $R_3 \rightarrow (-\frac{1}{4})R_3$, $R_4 \rightarrow R_4 - R_3$.

$$\left[\begin{array}{ccccc} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & -2 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Row op: $R_1 \rightarrow R_1 - 4R_3$.

$$\left[\begin{array}{ccccc} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Quick Determinant Tips

$\det(A)$ changes sign when rows are swapped; scaling a row by k multiplies $\det$ by k ; if two rows are identical or one is all zeros $\rightarrow \det(A) = 0$.

Example – Transformation That Is Not Linear

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $T(x_1, x_2) = x_1^2 + x_2$.

$T(\vec{x} + \vec{y}) = (x_1 + y_1)^2 + (x_2 + y_2) = x_1^2 + 2x_1y_1 + y_1^2 + x_2 + y_2 \neq T(\vec{x}) + T(\vec{y})$ (because of the $2x_1y_1$ term). Thus T fails additivity not linear.

Proof – Orthogonal Set Is a Subspace

Let $\vec{u} \in \mathbb{R}^2$ and define $S = \{\vec{x} \in \mathbb{R}^2 \mid \vec{x} \cdot \vec{u} = 0\}$.

- If $\vec{x}, \vec{y} \in S$, then $(\vec{x} + \vec{y}) \cdot \vec{u} = \vec{x} \cdot \vec{u} + \vec{y} \cdot \vec{u} = 0 + 0 = 0$ $\vec{x} + \vec{y} \in S$.

- If $k \in \mathbb{R}$ and $\vec{x} \in S$, then $(k\vec{x}) \cdot \vec{u} = k(\vec{x} \cdot \vec{u}) = k(0) = 0$ $k\vec{x} \in S$.

Hence S is closed under addition and scalar multiplication S is a subspace.

Problem 10(a) (Original Statement): Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Prove that if

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

then

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution. Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a + 2c & b + 2d \\ 2a + c & 2b + d \end{bmatrix}.$$

Setting $AB = 0$ gives

$$a + 2c = 0, \quad 2a + c = 0 \Rightarrow a = c = 0, \quad b + 2d = 0, \quad 2b + d = 0 \Rightarrow b = d = 0, \quad \text{so } B = 0.$$

Problem 10(b) (Original Statement): Let

$$C = \begin{bmatrix} 0 & 1 \\ c & 1 \end{bmatrix}.$$

Give the general form of a matrix D such that

$$CD = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution. Let $D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$CD = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ c & d \end{bmatrix}.$$

For $CD = 0$ we need $c = 0$ and $d = 0$. Hence the general form is

$$D = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

Problem 11 (Original Statement):

In $\mathbb{R}^2$, let

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Show that any vector

$$\vec{w} = \begin{bmatrix} a \\ b \end{bmatrix}$$

can be written as a linear combination $\vec{w} = r\vec{u} + s\vec{v}$.

Solution:

We need to find scalars r and s such that

$$r \begin{bmatrix} 1 \\ 2 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

This gives the system

$$\begin{cases} r + 2s = a, \\ 2r + s = b. \end{cases}$$

Multiply the first equation by 2:

$$2r + 4s = 2a$$

Subtract the second equation:

$$(2r + 4s) - (2r + s) = 2a - b \Rightarrow 3s = 2a - b \Rightarrow s = \frac{2a - b}{3}.$$

Substitute into $r + 2s = a$:

$$r + 2\left(\frac{2a - b}{3}\right) = a \Rightarrow 3r + 4a - 2b = 3a \Rightarrow 3r = -a + 2b \Rightarrow r = \frac{-a + 2b}{3}.$$

Hence

$$\begin{bmatrix} a \\ b \end{bmatrix} = \left(\frac{-a + 2b}{3}\right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \left(\frac{2a - b}{3}\right) \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(a) Is $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ invertible? Why or why not?

(b) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Find A^{-1} and solve $A\vec{x} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$.

Solution.

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 4 & 5 & 6 & | & 0 & 1 & 0 \\ 7 & 8 & 9 & | & 0 & 0 & 1 \end{bmatrix}$$

Row ops: $R_2 \rightarrow R_2 - 4R_1$, $R_3 \rightarrow R_3 - 7R_1$.

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & -3 & -6 & | & -4 & 1 & 0 \\ 0 & -6 & -12 & | & -7 & 0 & 1 \end{bmatrix}$$

Row op: $R_3 \rightarrow R_3 - 2R_2$.

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & -3 & -6 & | & -4 & 1 & 0 \\ 0 & 0 & 0 & | & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{not invertible.}$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 2 & 1 & 0 & | & 2 & -1 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

Row op: $R_2 \rightarrow R_2 - 2R_1$.

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & -1 & -2 & | & -2 & -1 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

Row ops: $R_2 \rightarrow -R_2$, $R_1 \rightarrow R_1 - R_3$.

$$\begin{bmatrix} 1 & 0 & -1 & | & 1 & 0 & 1 \\ 0 & 1 & 2 & | & 2 & -1 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

Row op: $R_3 \rightarrow R_3 - R_2$.

$$\begin{bmatrix} 1 & 0 & -1 & | & 1 & 0 & 1 \\ 0 & 1 & 2 & | & 2 & -1 & 0 \\ 0 & 0 & -1 & | & -2 & 1 & 1 \end{bmatrix}$$

Row op: $R_3 \rightarrow -R_3$.

$$\begin{bmatrix} 1 & 0 & -1 & | & 1 & 0 & 1 \\ 0 & 1 & 2 & | & 2 & -1 & 0 \\ 0 & 0 & 1 & | & 2 & -1 & -1 \end{bmatrix}$$

$$\text{Thus } A^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 1 & 2 \\ 2 & -1 & -1 \end{bmatrix}, \quad \vec{x} = A^{-1} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix}.$$

Problem 13 (Original Statement): Let $A = \begin{bmatrix} 1 & r \\ 2r & 1 \end{bmatrix}$.

- (a) Find $\det A$.
 (b) For which r is A invertible?
 (c) When invertible, give A^{-1} .

Solution.

$$\det A = (1)(1) - (2r)(r) = 1 - 2r^2.$$

$$\det A = 0 \Rightarrow 1 - 2r^2 = 0 \Rightarrow 2r^2 = 1 \Rightarrow r = \pm \frac{1}{\sqrt{2}}.$$

Thus A invertible for $r \neq \pm \frac{1}{\sqrt{2}}$.

$$A^{-1} = \frac{1}{1 - 2r^2} \begin{bmatrix} 1 & -r \\ -2r & 1 \end{bmatrix}.$$

Key Formulas and Facts

- Matrix Invertibility Tests:**
 - $\det(A) \neq 0 \iff A$ invertible
 - A invertible $\iff \text{rank}(A) = n \iff \text{rref}(A) = I$
- Inverse of 2×2 :** $A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- Linear Transformation:** $T(\mathbf{x}) = A\mathbf{x}$, T linear $\iff T(c\mathbf{x} + \mathbf{y}) = cT(\mathbf{x}) + T(\mathbf{y})$
- Composition:** $T_B \circ T_A = T_{BA}$
- Projection onto a line:** $\text{proj}_{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{x} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$
- Reflection across a line:** $\text{ref}_L(\mathbf{x}) = 2 \text{proj}_L(\mathbf{x}) - \mathbf{x}$
- Rotation matrix:** $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ preserves length.
- General solution of $A\mathbf{x} = \mathbf{b}$:** $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$

Quick Reference – Matrix Operations, Rank, Determinant & Solving $A\vec{x} = \vec{b}$

Matrix operations.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \quad A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}, \quad 2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

Multiplication (row $\times$ col): $AB = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$. Size rule: $(m \times n)(n \times p) \rightarrow (m \times p)$.
REF vs. RREF examples.

$$\text{REF} \begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 1 & 4 & | & 5 \\ 0 & 0 & 0 & | & 1 \end{bmatrix} \quad \text{RREF} \begin{bmatrix} 1 & 0 & -1 & | & 1 \\ 0 & 1 & 2 & | & 2 \\ 0 & 0 & 1 & | & -1 \end{bmatrix}$$

REF: zeros below pivots. RREF: pivots are 1 and the only nonzero in their columns.

Rank and nullity. $\text{rank}(A) = \# \text{ pivot cols}$, $\text{nullity}(A) = n - \text{rank}(A)$. If $\text{rank}([A \mid \vec{b}]) > \text{rank}(A) \Rightarrow$ no solution. If consistent: free vars $\Rightarrow$ infinitely many; no free vars $\Rightarrow$ unique.

Solving $A\vec{x} = \vec{b}$. 1) Form $[A \mid \vec{b}]$. 2) Row-reduce to RREF. 3) Pivots $\Rightarrow$ solved vars; nonpivots $\Rightarrow$ parameters. 4) $\vec{x} = \vec{x}_p + \vec{x}_h$ (particular + homogeneous).

Determinant essentials. Row swap $\Rightarrow$ sign flip; $R_i \leftarrow kR_i \Rightarrow \det \times k$; $R_i \leftarrow R_i + kR_j \Rightarrow \det$ unchanged; triangular U : $\det(U) = \prod u_{ii}$; square A : $\det(A) \neq 0 \iff \text{rank}(A) = n \iff \text{rref}(A) = I_n$ (invertible).

Examples. $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$, $\det \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = 0 \Rightarrow \text{rank} = 2 < 3$ (not invertible).

How to Solve $A\vec{x} = \vec{b}$ (Step-by-step)

Goal. Find all $\vec{x}$ with $A\vec{x} = \vec{b}$.

Recipe.

- Form the augmented matrix $[A \mid \vec{b}]$.
- Row-reduce $[A \mid \vec{b}]$ to **RREF** using elementary row ops (swap, scale a row, add multiple of a row).
- Check consistency:** any row $[0 \ 0 \ \dots \ 0 \mid c]$ with $c \neq 0$ means *no solution*.
- Identify **pivot columns** (leading 1's) and **free columns** (no pivot).
- Solve pivot variables in terms of the free variables (parameters $r, s, \dots$).
- Write the general solution as

$$\vec{x} = \vec{x}_p + \vec{x}_h, \quad \vec{x}_h = \text{all solutions to } A\vec{x} = \vec{0}.$$

Rank test (useful shortcut).

$$\text{rank}([A \mid \vec{b}]) > \text{rank}(A) \Rightarrow \text{no solution.}$$

If consistent:

free vars present $\Rightarrow$ infinitely many solutions, no free vars $\Rightarrow$ unique solution.

For square $n \times n$: unique solution $\iff \text{rank}(A) = n$.

Mini example (free variable).

$$\begin{bmatrix} 1 & 2 & 1 & | & 4 \\ 0 & 1 & 3 & | & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -5 & | & 0 \\ 0 & 1 & 3 & | & 2 \end{bmatrix}.$$

Let $x_3 = t$ (free). Then $x_2 = 2 - 3t$, $x_1 = 5t$.

$$\vec{x} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}}_{\vec{x}_p} + t \underbrace{\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}}_{\text{one basis vector of } \vec{x}_h}.$$

Mini example (no solution).

$$\begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 1 & 4 & | & 5 \\ 0 & 0 & 0 & | & 1 \end{bmatrix} \Rightarrow \text{row } [0 \ 0 \ 0 \mid 1] \Rightarrow \text{inconsistent (no solution).}$$

Problems 2 and 3 — Explanations and Solutions

Problem 2. Given the RREF

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

pivots are in x_1, x_2, x_3 ; x_4 is free. Rows give

$$x_1 + x_4 = 0, \quad x_2 + 2x_4 = 1, \quad x_3 + 3x_4 = 2.$$

Let $x_4 = r$. Then

$$x_1 = -r, \quad x_2 = 1 - 2r, \quad x_3 = 2 - 3r, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} + r \begin{bmatrix} -1 \\ -2 \\ -3 \\ 1 \end{bmatrix}.$$

Problem 3. System

$$\begin{cases} x_1 - x_3 = 1, \\ 2x_1 - 2x_2 = 1, \\ 3x_1 - 4x_2 + x_3 = a \end{cases} \implies \begin{bmatrix} 1 & 0 & -1 & | & 1 \\ 2 & -2 & 0 & | & 1 \\ 3 & -4 & 1 & | & a \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & | & 1 \\ 0 & 1 & -1 & | & \frac{1}{2} \\ 0 & 0 & 0 & | & a - 1 \end{bmatrix}.$$

Last row: $0 = a - 1$.

$a \neq 1 \Rightarrow$ no solution; $a = 1 \Rightarrow$ consistent with one free var $\Rightarrow$ infinitely many; never exactly one (missing a pivot). For $a = 1$:

$$x_1 - x_3 = 1, \quad x_2 - x_3 = \frac{1}{2}, \quad x_3 = r \Rightarrow x_1 = 1 + r, \quad x_2 = \frac{1}{2} + r,$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \\ 0 \end{bmatrix} + r \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Summary: (a) $a \neq 1$: no solution; (b) never exactly one; (c) $a = 1$: infinitely many (free var x_3).